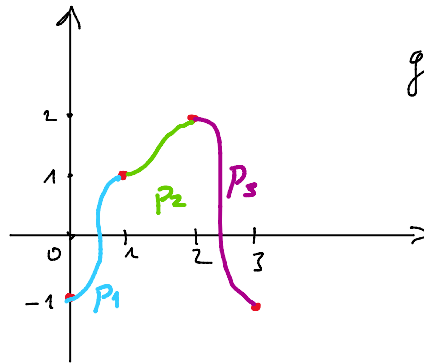


Ex.: Sa se determine spline-ul (cubic de clasa C^2) de Boor pentru nodurile: 0, 1, 2, 3 si valorile: -1, 1, 2, -1.

$$f(x) = \begin{cases} p_1(x), & x \in [0, 1] \\ p_2(x), & x \in [1, 2] \\ p_3(x), & x \in [2, 3] \end{cases}$$

$f \in C^2[0, 3]$ si
f interpolarea punctele



grad $p_i \leq 3$

$$p_i(x) = c_0 + c_1x + c_2x^2 + c_3x^3$$

SAU

$$p_i(x) = c_0 + c_1(x-x_0) + c_2(x-x_0)^2 + c_3(x-x_0)^3$$

$$\hat{f}(x) = \begin{cases} \underline{c_{10}} + \underline{c_{11}}(x-1) + \underline{c_{12}}(x-1)^2 + \underline{c_{13}}(x-1)^3, & x \in [0, 1] \\ \underline{c_{20}} + \underline{c_{21}}(x-1) + \underline{c_{22}}(x-1)^2 + \underline{c_{23}}(x-1)^3, & x \in [1, 2] \\ \underline{c_{30}} + \underline{c_{31}}(x-2) + \underline{c_{32}}(x-2)^2 + \underline{c_{33}}(x-2)^3, & x \in [2, 3] \end{cases}$$

$$f(0) = -1 \Rightarrow c_{10} - c_{11} + c_{12} - c_{13} = -1 \Rightarrow -c_{11} + c_{12} - c_{13} = -2 \quad (3)$$

$$f(1) = 1 \Rightarrow c_{10} = 1 = c_{20} \quad (1)$$

$$f(2) = 2 \Rightarrow c_{21} + c_{22} + c_{23} = 1, \quad c_{30} = 2$$

$$f(3) = -1 \Rightarrow c_{31} + c_{32} + c_{33} = -3 \quad (5)$$

$$p_1'(1) = p_2'(1) \Rightarrow c_{11} = c_{21}; \quad p_1''(1) = p_2''(1) \Rightarrow c_{12} = c_{22};$$

$$\text{de Boor} \Rightarrow p_1''(1) = p_2''(1) \Rightarrow c_{13} = c_{23}; \quad (2)$$

$$p_2'(2) = p_3'(2) \Rightarrow c_{21} + 2c_{22} + 3c_{23} = c_{31}$$

$$p_2''(2) = p_3''(2) \Rightarrow 2c_{22} + 6c_{23} = 2c_{32} \quad (4)$$

$$p_2'''(2) = p_3'''(2) \Rightarrow 6c_{23} = 6c_{33}$$

$$(1), (2) \Rightarrow c_{11} + c_{12} + c_{13} = 1 \quad | + \Rightarrow c_{12} = -\frac{1}{2} = c_{22}$$

$$(3) \Rightarrow -1 - c_{11} + c_{12} - c_{13} = -2$$

$$\Rightarrow c_{11} = \frac{3}{2} - c_{13} = 2$$

$$(4) \Rightarrow c_{31} = c_{11} + 2(-\frac{1}{2}) + 3c_{13} = \frac{1}{2} + 2c_{13} = -\frac{1}{2}$$

$$c_{32} = -\frac{1}{2} + 3c_{13} = -2$$

$$c_{33} = c_{13} = -\frac{1}{2}$$

$$(5) \Rightarrow \frac{1}{2} + 2c_{13} - \frac{1}{2} + 3c_{13} + c_{12} = -3 \Rightarrow c_{13} = -\frac{1}{2}$$

$$f(x) = \begin{cases} 1 + 2(x-1) - \frac{1}{2}(x-1)^2 - \frac{1}{2}(x-1)^3, & x \in [0, 1] \\ 1 + 2(x-1) - \frac{1}{2}(x-1)^2 - \frac{1}{2}(x-1)^3, & x \in [1, 2] \\ 2 - \frac{1}{2}(x-2) - 2(x-2)^2 - \frac{1}{2}(x-2)^3, & x \in [2, 3] \end{cases}$$

1) function c=splinecubic(nodes,values,type,der_values)
 %c=the coefficients of the cubic spline of the given type
 %examples of call:
 %c=splinecubic(nodes,values,'complete',[-1,1]) % f(a)=-1,f(b)=1
 %c=splinecubic(nodes,values,'naturale') % f'(a)=f'(b)=0
 %c=splinecubic(nodes,values,'derivate2',[1,2]) % f'(a)=1,f'(b)=2
 %c=splinecubic(nodes,values,'deBoor') % not-a-knot
 n=length(nodes);

$$p_i(x) = c_{i,0} + c_{i,1}(x-x_i) + c_{i,2}(x-x_i)^2 + c_{i,3}(x-x_i)^3, \quad i = \overline{1, n-1}$$

```

n=length(nodes);
c=[];
dx=diff(nodes)'; ddiv=diff(values)'/dx;
A=zeros(n); b=zeros(n,1);
for i=2:n-1
    A(i,i-1:i+1)=[dx(i), 2*(dx(i-1)+dx(i)), dx(i-1)];
    b(i)=3*(dx(i)*ddiv(i-1)+dx(i-1)*ddiv(i));
endfor
switch type
case 'complete'
    A(1,1)=1; A(n,n)=1;
    b(1)=der_values(1); b(n)=der_values(2);
case 'naturel'
    A(1,1:2)=[2 1]; A(n,n-1:n)=[1 2];
    b(1)=3*ddiv(1); b(n)=3*ddiv(n-1);
case 'derivate2'
    A(1,1:2)=[2 1]; A(n,n-1:n)=[1 2];
    b(1)=3*ddiv(1)-der_values(1)*dx(1)/2;
    b(n)=3*ddiv(n-1)+der_values(2)*dx(n-1)/2;
case 'deBoor'
    A(1,1:3)=[dx(2)^2 dx(2)^2-dx(1)^2 -dx(1)^2];
    b(1)=2*(ddiv(1)*dx(2)^2-ddiv(2)*dx(1)^2);
    A(n,n-2:n)=[dx(n-1)^2 dx(n-1)^2-dx(n-2)^2 -dx(n-2)^2];
    b(n)=2*(ddiv(n-2)*dx(n-1)^2-ddiv(n-1)*dx(n-2)^2);
endswitch
m=A\b
c(:,1)=values(1:n-1);
c(:,2)=m(1:n-1);
c(:,4)=(m(2:n)+m(1:n-1)-2*ddiv)/(dx.^2);
c(:,3)=(ddiv-m(1:n-1))/dx-c(:,4)*dx;
endfunction

```

$$dx = [1, 1, 1]; \quad ddiv = [2, 1, -3]$$

$$\bar{A} = [A | b] = \begin{bmatrix} 1 & 0 & -1 & 0 & 2 \\ 1 & 4 & 1 & 0 & 9 \\ 0 & 1 & 4 & 1 & -6 \\ 0 & 1 & 0 & -1 & 8 \end{bmatrix} \xrightarrow{L_2 - L_1} \begin{bmatrix} 1 & 0 & -1 & 0 & 2 \\ 0 & 4 & 2 & 0 & 7 \\ 0 & 0 & 4 & 2 & -14 \\ 0 & 0 & -1 & -1 & 6 \end{bmatrix} \xrightarrow{L_2 \leftrightarrow L_4} \begin{bmatrix} 1 & 0 & -1 & 0 & 2 \\ 0 & 4 & 2 & 0 & 7 \\ 0 & 0 & -1 & -1 & 6 \\ 0 & 0 & 4 & 2 & -14 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & -1 & 0 & 2 \\ 0 & 1 & 0 & -1 & 8 \\ 0 & 0 & 4 & 2 & -14 \\ 0 & 0 & 2 & 4 & -25 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 & 0 & 2 \\ 0 & 1 & 0 & -1 & 8 \\ 0 & 0 & 2 & 4 & -25 \\ 0 & 0 & 4 & 2 & -14 \end{bmatrix}$$

$$\xrightarrow{L_4 - 2L_3} \begin{bmatrix} 1 & 0 & -1 & 0 & 2 \\ 0 & 1 & 0 & -1 & 8 \\ 0 & 0 & 2 & 4 & -25 \\ 0 & 0 & 0 & -6 & 36 \end{bmatrix}$$

$$m = \begin{bmatrix} \frac{3}{2} \\ 2 \\ -\frac{1}{2} \\ -6 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ -1 \\ \frac{1}{2} \\ -6 \end{bmatrix} + \begin{bmatrix} \frac{3}{2} \\ 2 \\ -\frac{1}{2} \\ -6 \end{bmatrix} - 2 \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \\ -\frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix} - \begin{bmatrix} \frac{3}{2} \\ 2 \\ -\frac{1}{2} \end{bmatrix} + \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ -2 \end{bmatrix}$$

$$\begin{bmatrix} -1.0000 & 1.5000 & 1.0000 & -0.5000 \\ 1.0000 & 2.0000 & -0.5000 & -0.5000 \\ 2.0000 & -0.5000 & -2.0000 & -0.5000 \end{bmatrix}$$

$$f(x) = \begin{cases} -1 + \frac{3}{2}(x-0) + 1(x-0)^2 - \frac{1}{2}(x-0)^3, & x \in [0, 1] \\ 1 + 2(x-1) - \frac{1}{2}(x-1)^2 - \frac{1}{2}(x-1)^3, & x \in [1, 2] \\ 2 - \frac{1}{2}(x-2) - 2(x-2)^2 - \frac{1}{2}(x-2)^3, & x \in [2, 3] \end{cases}$$

$$III \quad (p_1 - p_2)(1) = (p_1 - p_2)'(1) = (p_1 - p_2)'''(1) = (p_1 - p_2)''(1) = 0$$

$$T_3(p_1 - p_2) = 0 \quad , \quad p_1 - p_2 = \text{pol. de grad} \leq 3 \Rightarrow p_1 - p_2 = T_3(p_1 - p_2) = 0.$$

$\hookrightarrow$ pol. Taylor d'ord. 3
 au point lui 1

$\Rightarrow p_1 = p_2.$

La fel, $p_2 = p_3.$

$$\text{Deci } f(x) = p_1(x), \quad x \in [0, 3].$$

$$\text{f pol. de grad} \leq 3 \quad \text{a.n.} \quad f(0) = -1, f(1) = 1, f(2) = 2, f(3) = -1$$

$\Rightarrow$ f pol. de interp. Lagr., de ord. 3

$$f(x) = \frac{(x-1)(x-2)(x-3)}{(0-1)(0-2)(0-3)} \cdot (-1) + \frac{x(x-2)(x-3)}{1(1-2)(1-3)} \cdot 1 +$$

$$+ \frac{x(x-1)(x-3)}{2(2-1)(2-3)} \cdot 2 + \frac{x(x-1)(x-2)}{3(3-1)(3-2)} \cdot (-1)$$

$$= \frac{1}{6}(x^3 - 6x^2 + 11x - 6) + \frac{1}{2}(x^3 - 5x^2 + 6x) - (x^3 - 4x^2 + 3x)$$

$$- \frac{1}{6}(x^3 - 3x^2 + 2x) = -\frac{1}{2}x^3 + x^2 + \frac{5}{2}x - 1, \quad x \in [0, 3].$$